```
{
  "project": {
    "name": "The Galactic Canvas",
    "type": "Interactive 3D Educational Website",
    "technology": [
      "HTML",
      "CSS",
      "JavaScript",
      "Three.js",
      "GLTFLoader"
    ],
    "deployment": "GitHub Pages",
    "theme": "Cinematic Interactive Space Journey",
    "goal": "Create a cinematic interactive journey from the Milky Way into the Solar System and then through the Sun and all eight planets."
  },

  "critical_instruction": {
    "existing_project": "Use the existing website as the starting point. Do not rebuild the entire project from scratch unless technically necessary.",
    "preserve": [
      "Existing Three.js architecture where practical",
      "Existing astronomical information",
      "Existing planet information",
      "Existing Learn More functionality",
      "Existing dark space visual language",
      "Responsive behavior",
      "Reduced-motion accessibility",
      "Existing navigation functionality"
    ],
    "priority": "VERY HIGH"
  },

  "asset_library": {
    "description": "The GitHub repository already contains the production 3D GLB assets inside assets/models/. These files must be used as the actual 3D objects in the website.",
    "base_path": "./assets/models/",
    "format": "GLB / glTF 2.0",
    "loader": "THREE.GLTFLoader",

    "models": {
      "milkyWay": "./assets/models/milky-way.glb",
      "solarSystem": "./assets/models/solar-system-animation.glb",
      "sun": "./assets/models/sun.glb",
      "mercury": "./assets/models/mercury.glb",
      "venus": "./assets/models/venus.glb",
      "earth": "./assets/models/earth.glb",
      "mars": "./assets/models/mars.glb",
```

```
      "jupiter": "./assets/models/jupiter.glb",
      "saturn": "./assets/models/saturn.glb",
      "uranus": "./assets/models/uranus.glb",
      "neptune": "./assets/models/neptune.glb"
    },

    "asset_rules": [
      "Use the supplied GLB models as the production models.",
      "Do not replace supplied GLB models with primitive spheres.",
      "Do not generate fake CSS planets.",
      "Do not use placeholder models in the final implementation.",
      "Load GLB files using Three.js GLTFLoader.",
      "Use relative paths so the website works correctly on GitHub Pages.",
      "Lazy-load heavy planet models when appropriate.",
      "Preload the next planet before the camera reaches it.",
      "Display a subtle loading state when a model is loading.",
      "Display a graceful fallback if a model fails to load.",
      "Do not expose raw GitHub URLs inside the application when relative repository paths
can be used."
    ]
  },

  "repository_structure": {
    "expected": [
      "index.html",
      "assets/models/milky-way.glb",
      "assets/models/solar-system-animation.glb",
      "assets/models/sun.glb",
      "assets/models/mercury.glb",
      "assets/models/venus.glb",
      "assets/models/earth.glb",
      "assets/models/mars.glb",
      "assets/models/jupiter.glb",
      "assets/models/saturn.glb",
      "assets/models/uranus.glb",
      "assets/models/neptune.glb"
    ],
    "optional": [
      "css/style.css",
      "js/main.js",
      "js/asset-loader.js",
      "js/camera-system.js",
      "js/scroll-controller.js",
      "js/planet-data.js"
    ]
  },

  "overall_experience": {
```

```
  "concept": "The user should feel like they are physically travelling through space rather than scrolling through ordinary webpage sections.",

  "journey": [
    "Milky Way Introduction",
    "Milky Way Information",
    "Space Jump / Nebula Transition",
    "Solar System",
    "Sun",
    "Mercury",
    "Venus",
    "Earth",
    "Mars",
    "Jupiter",
    "Saturn",
    "Uranus",
    "Neptune"
  ],

  "interaction": {
    "primary": "Scroll-driven cinematic camera movement",
    "secondary": [
      "Mouse interaction",
      "Touch interaction",
      "Planet selection",
      "Learn More",
      "Navigation",
      "Keyboard navigation"
    ]
  },

  "visual_style": [
    "cinematic",
    "immersive",
    "premium",
    "minimal",
    "scientific",
    "futuristic",
    "editorial",
    "deep space"
  ]
 },

 "design_system": {
  "background": "#03030A",
  "primary_text": "#EEF1FF",
  "secondary_text": "#A9B0D0",
  "accent": "#8EA2FF",
```

```
    "panel_background": "rgba(14,16,34,0.45)",
    "border": "rgba(160,175,255,0.18)",
    "heading_font": "Space Grotesk",
    "body_font": "Inter",
    "ui_style": "minimal cinematic glassmorphism",
    "border_radius": "14-18px",
    "visual_priority": "3D object first, typography second, UI third"
  },

  "core_3d_system": {
    "renderer": {
      "type": "THREE.WebGLRenderer",
      "antialias": true,
      "alpha": false,
      "pixelRatio": "Math.min(window.devicePixelRatio, 2)",
      "shadows": true
    },

    "scene": {
      "background": "#03030A"
    },

    "camera": {
      "type": "THREE.PerspectiveCamera",
      "smoothMovement": true,
      "continuousInterpolation": true,
      "dynamicTarget": true
    },

    "controls": {
      "desktop": "Allow subtle mouse interaction without fighting scroll-based camera control.",
      "mobile": "Support touch interaction.",
      "scroll": "Scroll controls the cinematic camera journey."
    }
  },

  "model_normalization": {
    "enabled": true,

    "purpose": "Different downloaded GLB files may have different scales, origins and
orientations. Normalize them automatically.",

    "process": [
      "Load the GLB.",
      "Calculate THREE.Box3 from the complete model.",
      "Calculate model size.",
      "Calculate model center.",
      "Calculate bounding sphere.",
```

```
    "Normalize visual scale according to the section requirements.",
    "Correct model position so the visual center aligns with the intended scene anchor.",
    "Preserve the model's internal materials and textures.",
    "Do not assume every model uses the same original scale."
  ],

  "important": [
    "Never hard-code one universal camera distance for all planets.",
    "Never assume the model origin is its visual center.",
    "Calculate camera distance from the normalized bounding sphere."
  ]
},

"page_01": {
  "name": "Milky Way Introduction",

  "purpose": "Create a strong cinematic opening.",

  "model": "./assets/models/milky-way.glb",

  "visual": {
    "description": "Use the Milky Way GLB as the main visual environment.",
    "solarSystem": "Do not prominently reveal the Solar System yet.",
    "composition": "Large-scale immersive galaxy environment."
  },

  "hero_text": {
    "title": "THE GALACTIC CANVAS",
    "subtitle": "A visual journey through our galaxy, solar system and planets.",
    "scroll_hint": "SCROLL TO EXPLORE"
  },

  "text_layout": {
    "position": "center",
    "background": "transparent",
    "avoidLargePanel": true
  },

  "animation": [
    "Slow galaxy movement",
    "Title fade-in",
    "Subheading delayed fade-in",
    "Subtle scale-up",
    "Scroll hint pulse"
  ]
},

"page_02": {
```

```
  "name": "The Milky Way",

  "purpose": "Explain the galaxy before entering the Solar System.",

  "model": "./assets/models/milky-way.glb",

  "camera": {
    "movement": "Slow cinematic movement through the galaxy.",
    "target": "Gradually identify the Solar System's region."
  },

  "headline": "THE MILKY WAY",

  "description": "A vast barred spiral galaxy containing billions of stars, including our Sun.",

  "information": [
    {
      "label": "GALAXY TYPE",
      "value": "BARRED SPIRAL"
    },
    {
      "label": "LOCATION",
      "value": "ORION ARM"
    },
    {
      "label": "SCALE",
      "value": "Approximately 100,000 light-years across"
    },
    {
      "label": "OUR HOME",
      "value": "The Solar System"
    }
  ],

  "text_behavior": {
    "style": "Editorial floating typography",
    "showOneAtATime": true,
    "background": "transparent or extremely subtle",
    "animation": "fade and translate"
  }
},

"page_03": {
  "name": "Space Jump Transition",

  "purpose": "Physically travel from the Milky Way toward the Solar System.",

  "sequence": [
```

```
      "Camera identifies the Solar System location.",
      "Galaxy particles gradually accelerate.",
      "Stars begin stretching toward the camera.",
      "Nebula particles appear.",
      "Camera accelerates forward.",
      "Create subtle hyperspace depth.",
      "Reduce acceleration.",
      "Reveal the Solar System."
    ],

    "effects": [
      "particle acceleration",
      "star streaks",
      "nebula clouds",
      "depth illusion",
      "subtle motion blur",
      "camera zoom",
      "smooth easing"
    ],

    "important": "This must feel like physical travel through space, not a simple fade
transition."
  },

  "page_04": {
    "name": "Solar System",

    "purpose": "Introduce the Solar System as the main interactive environment.",

    "model": "./assets/models/solar-system-animation.glb",

    "camera": {
      "startingShot": "medium-wide cinematic shot",
      "composition": "Balanced negative space.",
      "avoid": "Do not make the entire Solar System fill the screen."
    },

    "visual_positioning": {
      "important": "Optimize for visual storytelling rather than literal astronomical distance.",
      "rule": "Keep planets clearly separated and recognizable.",
      "avoid": [
        "Overlapping planets",
        "Tiny planets",
        "One giant planet dominating everything",
        "Flat 2D positioning"
      ]
    },
```

```
  "headline": "THE SOLAR SYSTEM",

  "subheading": "One star. Eight planets. Countless smaller worlds.",

  "information": [
    "ONE STAR",
    "EIGHT PLANETS",
    "BILLIONS OF SMALLER OBJECTS",
    "4.6 BILLION YEARS OLD"
  ],

  "interaction": [
    "Rotate the Solar System subtly.",
    "Hover over planets.",
    "Highlight selected planet.",
    "Click a planet to travel toward it.",
    "Scroll to travel toward the Sun.",
    "Allow navigation to individual planets."
  ]
},

"page_05": {
  "name": "Sun",

  "model": "./assets/models/sun.glb",

  "purpose": "Create a dedicated cinematic Sun experience.",

  "camera": {
    "shot": "Close-up / half-shot",
    "screenCoverage": "50-70%",
    "composition": "Sun positioned off-center with information in negative space."
  },

  "headline": "THE SUN",

  "subheading": "The star at the heart of our Solar System.",

  "data": [
    {
      "label": "DIAMETER",
      "value": "1,392,700 km"
    },
    {
      "label": "AGE",
      "value": "4.6 billion years"
    },
    {
```

```
      "label": "SURFACE TEMPERATURE",
      "value": "≈5,500 °C"
    },
    {
      "label": "CORE TEMPERATURE",
      "value": "≈15 million °C"
    }
  ],

  "visual": [
    "Solar surface movement if supported by asset",
    "Subtle glow",
    "Slow rotation",
    "Cinematic lighting"
  ]
},

"planet_system": {
  "general_rule": "Every planet must have its own visual identity and camera composition.",

  "critical_rules": [
    "Never use the same camera position for every planet.",
    "Never center every planet.",
    "Never use the same text position for every planet.",
    "Never use identical planet scale.",
    "Calculate camera distance from model bounds.",
    "Use negative space intelligently.",
    "Keep text away from important planetary details.",
    "Allow the planet to dominate the visual composition."
  ],

  "camera_behavior": {
    "entry": "Camera physically travels from the previous celestial body.",
    "arrival": "Slow cinematic deceleration.",
    "focus": "Camera targets the visual center of the planet.",
    "exit": "Camera begins moving toward the next destination before switching sections.",
    "interpolation": "Smooth lerp / damped interpolation.",
    "noAbruptCuts": true
  }
},

"planet_compositions": {

  "mercury": {
    "model": "./assets/models/mercury.glb",
    "camera": "Close angled shot",
    "planetPosition": "right",
    "textPosition": "left",
```

```
  "screenCoverage": "35-45%",
  "headline": "THE SWIFT MESSENGER",
  "visualFocus": [
    "craters",
    "rocky surface",
    "strong sunlight"
  ]
},

"venus": {
  "model": "./assets/models/venus.glb",
  "camera": "Medium close-up",
  "planetPosition": "left",
  "textPosition": "right",
  "headline": "A VEILED WORLD OF CRUSHING HEAT",
  "visualFocus": [
    "thick atmosphere",
    "clouds",
    "yellow-orange atmosphere"
  ]
},

"earth": {
  "model": "./assets/models/earth.glb",
  "camera": "Cinematic half-planet shot",
  "planetPosition": "right",
  "textPosition": "left",
  "headline": "THE ONLY KNOWN WORLD WITH LIFE",
  "visualFocus": [
    "blue oceans",
    "clouds",
    "atmosphere",
    "Moon if included in the asset"
  ]
},

"mars": {
  "model": "./assets/models/mars.glb",
  "camera": "Angled close-up",
  "planetPosition": "left",
  "textPosition": "right",
  "headline": "THE RUSTY RED FRONTIER",
  "visualFocus": [
    "red surface",
    "canyons",
    "mountains",
    "dust"
  ]
```

```
},

"jupiter": {
  "model": "./assets/models/jupiter.glb",
  "camera": "Medium-wide cinematic shot",
  "planetPosition": "right",
  "textPosition": "left",
  "headline": "KING OF THE PLANETS",
  "visualFocus": [
    "Great Red Spot",
    "cloud bands",
    "large scale",
    "moons if included"
  ]
},

"saturn": {
  "model": "./assets/models/saturn.glb",
  "camera": "Hero cinematic shot",
  "planetPosition": "center-left",
  "textPosition": "right",
  "headline": "THE RINGED JEWEL",
  "visualFocus": [
    "rings",
    "planetary atmosphere",
    "dramatic lighting"
  ],
  "composition": "Diagonal composition with the rings extending across the frame."
},

"uranus": {
  "model": "./assets/models/uranus.glb",
  "camera": "Medium close-up",
  "planetPosition": "right",
  "textPosition": "left",
  "headline": "THE TILTED ICE GIANT",
  "visualFocus": [
    "cyan color",
    "axial tilt",
    "subtle rings"
  ]
},

"neptune": {
  "model": "./assets/models/neptune.glb",
  "camera": "Distant cinematic close-up",
  "planetPosition": "left",
  "textPosition": "right",
```

```
    "headline": "THE DISTANT WINDSWEPT BLUE",
    "visualFocus": [
      "deep blue atmosphere",
      "storm systems",
      "dark space"
    ]
  }
 },

 "integrated_information_design": {
  "critical_change": "Do not make the planet information look like a separate rectangular website card.",

  "core_rule": "Typography and 3D objects must feel like one cinematic composition.",

  "default": {
   "background": "Transparent",
   "panel": "Only use a subtle glass background when readability requires it.",
   "border": "Minimal or none",
   "shadow": "Very subtle",
   "position": "Use negative space around the planet.",
   "maximumWidth": "360px"
  },

  "layout": {
   "possiblePositions": [
    "upper-left",
    "middle-left",
    "lower-left",
    "upper-right",
    "middle-right",
    "lower-right"
   ],

   "dynamicPositioning": true,

   "rules": [
    "Determine which side of the screen has the most negative space.",
    "Place information on the opposite side of the planet.",
    "Never cover the important visual area of the planet.",
    "Maintain a minimum visual gap between text and planet.",
    "Adjust text position dynamically based on camera composition.",
    "Keep the composition balanced."
   ]
  },

  "typography": {
   "headline": {
```

```
      "style": "Large editorial typography",
      "size": "clamp(2rem, 5vw, 5rem)",
      "weight": 700,
      "lineHeight": 0.95
    },

    "description": {
      "style": "Short editorial sentence",
      "maximumLines": 3
    },

    "facts": {
      "style": "Small floating metadata",
      "layout": "horizontal or vertical depending on available space"
    },

    "labels": {
      "style": "Uppercase",
      "letterSpacing": "0.12em"
    }
  },

  "visual_connection": {
    "enabled": true,
    "useSubtleConnectorLine": true,
    "useOrbitLine": false,

    "rules": [
      "A subtle line may connect a fact to a relevant planetary region.",
      "Do not use excessive decorative lines.",
      "Do not obscure the planet.",
      "Connector lines should animate softly."
    ]
  }
},

"learn_more_system": {
  "enabled": true,

  "defaultView": {
    "content": "Short cinematic facts",
    "maximumFacts": 3
  },

  "expandedView": {
    "style": "Expanded educational overlay",
    "animation": "Smooth fade and scale",
    "background": "Subtle dark glass",
```

```
    "closeButton": true
   },

   "important": "The Learn More interface must not permanently cover the planet."
  },

  "scroll_system": {
   "primaryFunction": "Scroll controls the camera journey.",

   "behavior": [
    "Each section has a defined camera start state.",
    "Each section has a defined camera end state.",
    "Camera position interpolates continuously.",
    "Camera target interpolates continuously.",
    "Object scale interpolates when required.",
    "Lighting transitions between scenes.",
    "Text fades and moves during transitions.",
    "The next model begins loading before the transition."
   ],

   "easing": "Smooth cinematic easing",

   "avoid": [
    "Abrupt camera jumps",
    "Instant model switching",
    "Hard fades between planets",
    "Sudden text changes"
   ]
  },

  "text_animation": {
   "entry": {
    "opacity": "0 to 1",
    "translateY": "30px to 0",
    "duration": "cinematic"
   },

   "exit": {
    "opacity": "1 to 0",
    "translateY": "0 to -20px"
   },

   "rule": "Text should enter as part of the camera composition, not as a generic webpage
section."
  },

  "navigation": {
   "style": "Minimal floating bottom navigation",
```

```
    "sections": [
      "Galaxy",
      "Milky Way",
      "Solar System",
      "Sun",
      "Mercury",
      "Venus",
      "Earth",
      "Mars",
      "Jupiter",
      "Saturn",
      "Uranus",
      "Neptune"
    ],

    "behavior": [
      "Highlight active section.",
      "Smooth scroll to selected section.",
      "Keep navigation unobtrusive.",
      "Allow keyboard navigation."
    ]
  },

  "responsive_design": {

    "desktop": {
      "experience": "Full-screen cinematic 3D",
      "text": "Integrated around the planet",
      "particles": "Full quality"
    },

    "tablet": {
      "reduceParticleCount": true,
      "reduceModelComplexityIfNecessary": true,
      "maintainReadableTypography": true
    },

    "mobile": {
      "reduceParticleDensity": true,
      "reduce3DScale": true,
      "textLayout": "Place information beside the planet when possible and below the planet when necessary.",
      "maintainTouchInteraction": true,
      "maintainReadableTypography": true,
      "avoidHorizontalOverflow": true
    }
  },
```

```
  "performance": {
    "limitPixelRatio": 2,
    "lazyLoadHeavyAssets": true,
    "preloadNextAsset": true,
    "reduceParticlesOnMobile": true,
    "avoidUnnecessaryDOMAnimation": true,
    "disposeUnusedResources": true,
    "maintainSmoothScrolling": true,

    "loading": {
      "showProgress": true,
      "style": "Minimal cinematic loader",
      "avoid": "Large traditional loading screen"
    }
  },

  "accessibility": {
    "keyboardNavigation": true,
    "reducedMotion": true,
    "manualReducedMotionToggle": true,
    "visibleFocusStates": true,
    "ariaLabels": true,
    "sufficientTextContrast": true
  },

  "error_handling": {
    "missingModel": {
      "behavior": "Show a subtle error message and continue the experience where possible."
    },

    "loadingFailure": {
      "behavior": "Do not crash the entire website because one GLB fails to load."
    },

    "githubPages": {
      "rule": "Use relative paths and ensure asset loading works when hosted from GitHub
Pages."
    }
  },

  "important_visual_rules": {
    "do": [
      "Make the 3D model the visual hero.",
      "Use cinematic negative space.",
      "Give every planet a unique composition.",
      "Use different camera angles.",
      "Use different text positions.",
```

```
      "Use smooth physical camera travel.",
      "Integrate typography with the 3D scene.",
      "Use supplied GLB assets.",
      "Normalize GLB scale automatically.",
      "Keep the interface minimal."
    ],

    "doNot": [
      "Do not use generic placeholder spheres.",
      "Do not put every planet in the center.",
      "Do not use the same camera position for every planet.",
      "Do not put all text into large rectangular boxes.",
      "Do not cover planets with information panels.",
      "Do not make the website look like a normal webpage with a 3D model behind it.",
      "Do not use abrupt scene changes.",
      "Do not make all planets visually identical in scale.",
      "Do not overload the screen with statistics.",
      "Do not destroy the existing educational content."
    ]
  },

  "final_experience": {
    "description": "The finished website should feel like an interactive space documentary.",

    "narrative": [
      "The user begins inside the Milky Way.",
      "The camera travels through the galaxy.",
      "The Solar System is discovered.",
      "The user approaches the Sun.",
      "The journey continues planet by planet.",
      "Each planet receives its own cinematic composition.",
      "Information appears naturally within the composition.",
      "The user can interact and learn more.",
      "The journey ends at Neptune."
    ],

    "mostImportantRule": "Never make the website feel like a static webpage placed on top of a 3D model. The 3D environment, camera movement, scroll, lighting, typography and information must work together as one continuous cinematic narrative."
  },

  "implementation_requirements": {
    "deliverable": "Complete working website",

    "mustInclude": [
      "HTML",
      "CSS",
      "JavaScript",
```

```
      "Three.js",
      "GLTFLoader",
      "Supplied GLB assets",
      "Scroll-based camera system",
      "Dynamic model positioning",
      "Dynamic model scaling",
      "Planet-specific camera compositions",
      "Integrated typography",
      "Planet navigation",
      "Interactive planets",
      "Learn More functionality",
      "Responsive design",
      "Reduced motion mode",
      "Loading states",
      "Error handling"
    ],

    "codeQuality": [
      "Well structured",
      "Commented",
      "Easy to modify",
      "Modular where practical",
      "Avoid unnecessary dependencies",
      "Use clean relative asset paths",
      "Do not duplicate large blocks of code unnecessarily"
    ]
  },

  "finalInstructionToAI": {
    "message": "Modify the existing website rather than creating a generic new website. The repository already contains the real GLB assets in assets/models/. Load and use those models. Focus especially on cinematic planet positioning, unique camera compositions, smooth scroll-driven travel, and typography that is visually integrated into the 3D scene rather than displayed in large separate boxes. The final result must feel like a premium interactive space documentary."
  }
}
```